

Balaji Murugaiya

Backend Software Engineer

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PROFESSIONAL SUMMARY

Backend Software Engineer with **5 years** of experience building and owning distributed identity, authentication, multi-tenant storage, and cloud platforms supporting up to **100M users** and approximately **1M requests/minute**. Delivered a zero-downtime migration of approximately **10M identifiers**, improved cache eviction by **100x**, and identified a projected **40% infrastructure cost reduction**.

KEY HIGHLIGHTS

- **Platform-scale backend and identity engineering:** Built an OAuth2 authorization platform supporting **3,000 clients** and **100M users**, and led OIDC-based SSO enablement for **320 on-prem clients**.
- **Production performance and cost optimization:** Delivered a zero-downtime migration of approximately **10M identifiers**, achieved **100x faster** cache eviction, and proposed approximately **\$8,000/year** in infrastructure savings.

CORE COMPETENCIES & TECHNOLOGIES

Core Competencies: Backend & Platform Engineering, Distributed Systems, Identity & Access Management, Cloud Infrastructure, Performance Optimization, Production Incident Response, Cost Optimization

Languages & Frameworks: Java, Spring Boot, gRPC, REST, Jersey Framework

Distributed Systems: Kafka, SQS, asynchronous messaging, idempotent consumers, eventual consistency, cache design

Databases & Caching: MySQL, DynamoDB, MongoDB, DocumentDB, Redis, Amazon Aurora RDS

Cloud & DevOps: AWS (ECS, EKS, RDS, SQS, S3, IAM), Kubernetes, Docker, Terraform, ArgoCD, Argo Rollouts, Jenkins

Security & Observability: OAuth2, OIDC, JWT/JWKS, PKCE, API key management, GDPR data lifecycle, Grafana, Elastalert, Prometheus, OpenTelemetry, distributed tracing

Quality & AI Tools: JUnit, Testcontainers, JaCoCo, static analysis, CI/CD, Cursor, Claude Code, MCP integrations

WORK EXPERIENCE

Freshworks | Bangalore, India

Senior Software Engineer - Entity Store Team

Apr 2025 - May 2026

- Independently designed and implemented a collision-free **53-bit ID generation scheme** to migrate **~10M Freshservice production identifiers** under a hard cutover deadline, ensuring JavaScript safe integer compatibility with **zero downtime**.
- Redesigned Formserv derived-form cache from async SQS-based push invalidation to a lazy pull model, eliminating stale field propagation across reference-derived form relationships and achieving **100x faster cache eviction**.
- Analyzed RDS utilization across **6 shards (US) ($\leq 20\%$ memory/CPU, ~10 read IOPS vs 40,000 supported)** and proposed right-sizing from db.r8g.xlarge to db.r8g.large - projected **~40% cost reduction (~\$8,000/year savings)**.
- Owned RCA for a **P1 incident (14-min degradation)** caused by incorrect rate limit configuration; implemented class-level preventive alerting to eliminate recurrence.

Senior Software Engineer - FreshID Team

Jan 2025 - Apr 2025

- Led end-to-end design and implementation of OIDC integration into the existing OAuth server, enabling SSO for **320 on-prem clients** (across a **3,000-client OAuth platform**) as part of the Device42 acquisition - delivered against a hard acquisition timeline.
- Prevented Redis memory exhaustion by identifying **~70,000 non-TTL session keys (700 MB)** through heap dump analysis and scripting bulk cleanup before LRU eviction impacted active user sessions.
- Proposed migrating **6 services across 5 regions (US, EU, Australia, India, and Middle East)** and **5 Kubernetes namespaces** to Argo Rollouts to replace an existing **150-step manual canary deployment process** with automated progressive delivery.

Software Engineer - FreshID Team

Apr 2022 - Dec 2024

- Built an OAuth2 Authorization Server using Spring Authorization Server, onboarding **3,000 OAuth clients** and handling **~10,000 req/min** for third-party Freshworks API integrations on a platform serving **100M users**.
- Implemented application performance monitoring using Grafana and Elastalert across the FreshID platform (**100M users, ~1M req/min**), establishing **P0 detection under 1 min** and **P1 detection under 5 mins** - previously undetectable without inbound customer tickets.
- Built gRPC-based chatbot integration for FreshID enabling L1 support teams to perform CRUD operations autonomously, deflecting **~20% of L2 support tickets**.

Graduate Trainee - Software Engineering

Aug 2021 - Apr 2022

Software Engineering Intern

Mar 2021 - Jun 2021

PROJECTS

Passkey Authentication (WebAuthn) | Internal Hackathon Prototype

- Prototyped WebAuthn passkey login to reduce password-related phishing and credential risk and avoid the setup overhead of SSO.
- Implemented tenant-level enablement and per-user passkey enrollment, storing WebAuthn public-key credential metadata and supporting platform authenticators such as fingerprint/Face ID; demonstrated an end-to-end passwordless login flow.

ACHIEVEMENTS

- Finalist, Smart India Hackathon 2019 - **Top 1% of 500,000 participants**, Ministry of HRD, Government of India.

EDUCATION

SSN College of Engineering | Chennai, India | B.E., Electronics & Communication | CGPA: 7.83/10 | 2017 - 2021